

Detection of Externally-Influenced Routing via Inverse Markov-Chain Methods

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MEng Design Engineering — Master’s Project

May 29, 2026

Abstract

Urban traffic increasingly carries the fingerprint of external routing influence: satellite-navigation systems and live traffic services reroute their users around congestion, producing observed traffic patterns that no longer reflect drivers’ intrinsic route preferences. Recovering those intrinsic preferences from partial network observations is the inverse problem of (author?) [1], who introduced a softmax-parametric Markov-chain inverter that fits a transition kernel to observed stationary distributions and hitting rates at a small subset of observation states. We extend that framework with three contributions that target real-world applicability. First, we wire SUMO-derived edge attributes (road class, speed, lanes, turn geometry) into the inverter as global features and show, across a five-seed SUMO sweep, that they robustly improve *per-row chain recovery* — the stacked Pearson correlation between fitted and empirical regime contrasts rises on all five seeds (mean $+0.050 \pm 0.023$, paired $p \approx 0.008$) — whereas the aggregate detection-AUC lift is seed-dependent and not statistically robust ($+0.024 \pm 0.026$, reversing on one of five seeds). On the representative seed the detector reaches $\text{AUC} = 0.708$ at $T=40$ against an empirically verified 1-step Markov data ceiling of 0.801. Second, we settle whether a parametric 2-step extension would close the remaining gap by comparing a 1-step empirical detector to a 2-step empirical detector built from the same trajectories: the 2-step *underperforms*, refuting higher-order memory as the missing lever and identifying the chosen 1-step softmax family as the binding model-class constraint. Third, we port the entire pipeline to a real-world urban Automatic-Vehicle-Identification dataset from Xuancheng, China [2] ($\sim 10.5\text{M}$ reconstructed vehicle trips over April 2023), with an OD-rebalancing extension that controls for the population-mix confound absent in simulated data. The OD-rebalancing reveals that approximately three-quarters of the apparent rush-vs-off-peak chain difference in raw pooled data is attributable to OD-mix between commuter and discretionary trip populations, not to routing-choice differences: the empirical 1-step Markov ceiling falls from $\text{AUC} = 0.573$ on the un-matched pool to $\text{AUC} = 0.518$ on the OD-matched pool. After controlling for this confound, the within-OD routing contrast is only 0.018 above chance — below the detectability threshold of the framework under the chosen behavioural proxy. Across all three validation tiers, methodology and pipeline remain identical; only the data changes. The resulting work establishes both an upper bound (synthetic reproduction within ~ 0.05 RMAE of the original paper) and an honest lower bound on what 1-step Markov inverse methods can achieve on real urban traffic without ground-truth rerouting labels.

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1 Notation

A consolidated reference for the mathematical objects used throughout this report. Notation follows (author?) [1] where it does, and is extended in section 4 where required.

Table 1: Key symbols.

Symbol	Meaning
$\mathcal{X}, n$	Full state space (directed road edges); $n = \mathcal{X} $.
$X_o \subset \mathcal{X}$	Subset of <i>observed</i> states at which counts are available.
$\text{adj}(x)$	Set of legal successors of state x (out-neighbours in the directed graph).
$p_T(x' x)$	One-step transition probability from x to x' ; row x of the kernel P_T .
$p_I(x)$	Initial / teleport probability on $\mathcal{X}$; row vector.
$\beta \in [0, 1]$	Continuation probability per step; together with p_I defines the stationary mixture $P = \beta P_T + (1 - \beta) \mathbf{1} p_I^\top$.
π	Stationary distribution of P ; obtained from $\pi^\top P = \pi^\top$, $\mathbf{1}^\top \pi = 1$.
$f(x)$	Empirical visit count (or normalised frequency) at x . Observed only for $x \in X_o$.
$g(x, x')$	Empirical (or analytical) <i>hitting rate</i> from x to x' , a discounted first-passage probability on $X_o \times X_o$.
$\phi_T(x') \in \mathbb{R}^{d_T}$	Destination-state feature vector (section 4.3).
$\psi(x, x') \in \mathbb{R}^{d_\psi}$	Directed-edge feature vector.
$\nu \in \mathbb{R}^n$	Per-state base parameters of p_I .
$\omega^{\text{loc}} \in \mathbb{R}^E$	Per-edge local weights of p_T , with $E = \sum_x \text{adj}(x) $ the number of directed edges.
$\omega^{\text{gl}_1} \in \mathbb{R}^{d_T}$	Shared weights on destination features ϕ_T (paper Eq. 17).
$\omega^{\text{gl}_2} \in \mathbb{R}^{d_\psi}$	Shared weights on edge features ψ (paper Eq. 17).
θ	Full parameter vector $[\nu; \omega^{\text{loc}}; \omega^{\text{gl}_1}; \omega^{\text{gl}_2}] \in \mathbb{R}^d$, $d = n + E + d_T + d_\psi$.
λ	Tikhonov regularisation strength on $\ \theta\ ^2$.
$\gamma \in [0, 1]$	Mixing weight between the stationary-match loss L_d and the hitting-rate loss L_h .
$\Lambda(\tau)$	Log-likelihood-ratio score of a test trajectory τ under two fitted chains (section 4).
AUC	Area under the ROC curve of the LR score against labelled regime test trajectories.
$\text{RMAE}(\hat{\pi}, \pi)$	Relative mean absolute error of recovered stationary mass at unobserved states.

Throughout, “regime” refers to one of two distinguishable driver populations whose chains we fit and compare. In controlled simulation this is intrinsic vs. sat-navigation-influenced; in real-world data it is a behavioural proxy (e.g. rush-hour vs. off-peak). The contrast $\Delta_{\text{fit}}(x, \cdot) = p_T^{(B)}(x, \cdot) - p_T^{(A)}(x, \cdot)$ between two fitted chains is the central quantity assessed against the empirical ground-truth contrast $\Delta_{\text{emp}}(x, \cdot)$.

2 Introduction

2.1 Context: urban traffic and the external-influence problem

The widespread adoption of satellite-navigation applications such as Google Maps, Waze, and TomTom has structurally altered how vehicles use urban road networks. These systems no longer merely report a single shortest path: they reroute users in real time around observed congestion, redistributing traffic across alternative arteries on time-scales of minutes to hours. The consequence is that any traffic observation aggregating across modern drivers reflects a

mixture of two distinct routing rules: the intrinsic preferences a driver would express in the absence of guidance (habit, scenic preference, fuel-cost optimisation, familiarity), and the external recommendations of a sat-navigation system reacting to live network state.

This mixture creates a problem for downstream traffic models. Demand estimation, route-choice inference, transport-policy evaluation, and infrastructure-investment decisions all rely on revealed-preference data that is assumed to encode *driver* preferences. When a sizeable fraction of observed trips are guided by an external recommender that itself responds to congestion (and thereby to other guided trips), the revealed data encodes a partially synthetic, congestion-coupled signal that the underlying driver population would not have produced unaided. Identifying and filtering this externally-influenced traffic, or at minimum quantifying its presence, has therefore become a methodological prerequisite for any inference about driver behaviour in modern urban environments.

2.2 Inverse Markov-chain methods and the Morimura framework

A long line of work models route choice as a Markov chain over road network edges, in which each driver, conditional on their current edge, selects a next edge with probabilities determined by local and global features of the network. (author?) [1] introduced an *inverse* formulation that fits the parameters of such a chain from only partial observations of the network — specifically, from vehicle counts at a small subset of observation stations X_o and optionally from pairwise hitting rates between those stations. Their softmax-parametric chain admits an analytic gradient of the regularised log-ratio loss, enabling reliable fitting on networks of several thousand edges with 20–90% observation coverage.

The Morimura framework is well validated on synthetic data and was applied by its authors to one real-world traffic case study in Nairobi. It has not, however, been adapted to the specific problem of detecting externally-influenced routing as distinct from intrinsic preferences: the original work treats the inverse problem as one of *recovering* a single chain, not of *distinguishing* two coexisting routing regimes within the same network.

2.3 Contributions

This dissertation extends the Morimura inverse-Markov-chain method into a three-tier detection pipeline for externally-influenced routing, validates each tier honestly, and quantifies where the framework’s limits actually lie. Specifically:

1. **Reproduction of the Morimura §6.1 synthetic benchmark.** We re-implement the inverter, derive and finite-difference-verify the analytic gradients (including the omega-global gradient extensions for shared destination and edge features that the paper presents but does not derive in full), and reproduce the headline RMAE curve in Fig. 2A of [1] within ~ 0.05 across observation fractions $X_o/n \in [0.10, 0.90]$. This locks in the upstream implementation as a faithful reference.
2. **Simulated-tier detector built on SUMO microscopic traffic simulation.** We construct a bigger-bbox SUMO scenario in which an identical population of vehicles is simulated twice — once with no sat-navigation rerouting (*intrinsic* regime) and once with full sat-navigation rerouting (*sat-nav* regime) — and fit a Morimura inverter to each regime’s partial observations. The fitted chains feed a per-trajectory log-likelihood-ratio detector for which the area under the ROC curve at $T = 20$ test transitions reaches $\text{AUC} = 0.644$ when the inverter is augmented with real SUMO edge features, against an empirically verified 1-step Markov data ceiling of 0.685 (closing 77.5% of the recoverable gap). A trajectory-length sweep lifts the ceiling and the detector together to $\text{AUC} = 0.708$

at $T = 40$ against a ceiling of 0.801.

3. **An honest model-class diagnostic.** The remaining gap to the 0.801 ceiling could in principle be closed by extending the inverter to a 2-step Markov family (modelling the choice of the next edge given the current *and previous* edge). We test this by constructing a 2-step *empirical* detector — counting $(x_{t-1}, x_t) \rightarrow x_{t+1}$ triples on the same trajectories — and comparing its AUC to the 1-step empirical ceiling. The 2-step detector *underperforms* ($\text{AUC}_{2\text{-step}} = 0.666 < \text{AUC}_{1\text{-step}} = 0.685$), refuting higher-order memory as the missing lever and identifying the 1-step softmax family itself as the binding model-class constraint. This is a publishable negative result that forecloses an otherwise-tempting line of work.
4. **Real-world validation tier on the Xuancheng AVI dataset.** We port the entire pipeline to a real-world urban traffic dataset of ~ 10.5 million reconstructed vehicle trips over April 2023 [2]. Because no real dataset comes with sat-navigation adoption labels, we use a time-of-day behavioural proxy (rush-hour vs. off-peak driving) and introduce an OD-rebalancing extension to the loader that subsamples each regime to share the same (origin zone, destination zone) distribution, removing the population-mix confound that distinguishes any real-world proxy contrast from a SUMO-style controlled comparison.
5. **An end-to-end pipeline, reusable across tiers.** A single CLI controls all three validation tiers via a `-dataset {sumo,xuancheng}` flag. The inverter, feature extractor, gradient implementation, and contrast diagnostics are unchanged across tiers; only the loader, the regime split, and the per-tier honest caveats differ. The artefact is in the repository at <https://github.com/placeholder>.

2.4 Roadmap

section 3 situates the work in relevant prior literature. section 4 presents the mathematical framework and the three extensions (real features, 2-step diagnostic, OD-rebalancing) in detail. sections 5 to 7 present the three validation tiers in turn, each one with its own honest discussion of what does and does not transfer. section 8 draws cross-tier conclusions and limitations. section 9 addresses project management and personal-development reflection per the module’s *Reflection* learning outcome [3].

3 Background and Related Work

3.1 Inverse problems on Markov chains

A Markov chain is a stochastic process in which the next state depends only on the current state. The *forward* problem asks, given a transition kernel P_T , what statistics (stationary distribution, hitting times, expected occupation, etc.) does the chain produce? The *inverse* problem reverses this: given partial statistics observed on the chain, recover the transition kernel itself. Inverse problems on Markov chains arise in transport modelling [4], biological systems [5], and recommender systems [6], and are typically ill-posed under sparse observation because many distinct kernels produce identical low-order statistics. The usual remedy is a parametric restriction: rather than fit $P_T \in \mathbb{R}^{n \times n}$ directly, one fits a parameter vector $\theta \in \mathbb{R}^d$ with $d \ll n^2$ such that the parametric form $P_T(\theta)$ is constrained to lie in a manifold of plausible chains. The choice of parametric family is then load-bearing: the wrong family caps recovery quality regardless of optimisation effort.

3.2 Route choice as a softmax-parametric Markov chain

In the route-choice setting the state space $\mathcal{X}$ is the set of directed road edges and the kernel $p_T(x' | x)$ is the probability that a vehicle currently on edge x moves next to edge x' . Multinomial-

logit route-choice models [4, 7] parametrise the transition probability as a softmax over the legal successors of x :

$$p_T(x' | x) = \frac{\exp(s(x, x'))}{\sum_{x'' \in \text{adj}(x)} \exp(s(x, x''))}, \quad (1)$$

where $s(x, x')$ is a real-valued score, typically a linear combination of features of the edge (x, x') and of the destination state x' . This formulation has two important properties for inverse problems. First, it is parameter-efficient: one global feature shared across many states adds one parameter, regardless of network size. Second, its gradient is analytic in s , which combined with the chain-rule propagation through stationary computations and hitting-rate solves makes $\nabla_\theta \log \pi$ and $\nabla_\theta \log g$ tractable to derive in closed form.

3.3 The Morimura inverse-MC framework

(author?) [1] apply the softmax form of eq. (1) to the partial-observation regime in which the modeller observes only a small subset $X_o \subset \mathcal{X}$. They consider two empirical observation types:

- **Empirical stationary distribution f at X_o .** A count of vehicle passages at each station; effectively a sample of $\pi(x)$ for $x \in X_o$.
- **Empirical hitting rates $g(x, x')$ on $X_o \times X_o$.** The fraction of vehicles observed at x that subsequently visit x' discounted by their travel time; effectively a sample of the β -discounted first-passage probability.

They define a regularised loss that mixes a log-ratio match on f (via the stationary distribution of the parametric chain) and a log-error match on g (via the hitting-rate matrix of the parametric chain), weighted by a mixing parameter γ . Their key contribution is the analytic derivation of the gradient of this loss with respect to θ , enabling optimisation via L-BFGS-B or, alternatively, the natural-gradient descent the paper introduces in §4.1 as the optimiser of choice when the loss landscape is ill-conditioned.

The paper validates the framework synthetically on a $n = 100$ network (their §6.1) and applies it to a 1497-link Nairobi minibus network as a real-world demonstration (§6.2). Both validations report favourable RMAE on the recovered stationary distribution. Two limitations of this validation are directly relevant to our work. First, the synthetic experiment uses analytical (i.e. noise-free) hitting rates g computed from the true chain; on noisy real-world data the empirical g is overwhelmingly sparse, an effect documented in section 6.6. Second, the real-world Nairobi case is not framed as a regime-detection problem: the authors do not split the data into two coexisting routing populations and ask whether the chain recovery distinguishes them.

3.4 Re-routing detection and external traffic influence

Detecting externally-influenced rerouting is a relatively recent research direction. (author?) [8] proposed a junction-level anomaly-detection approach in which deviations from a learned turning-distribution baseline serve as evidence of unusual routing behaviour; their setup is restricted to snapshot-level (population-level) detection rather than per-trajectory. Active-routing-game analyses [9, 10] model the equilibrium effects of sat-navigation adoption on aggregate network performance, but assume a known navigation rule and focus on welfare bounds rather than detection. Closest in spirit to this work is the line of route-choice research that combines partially observed counts with route-fragment likelihoods [11]: this addresses estimation under partial observation but does not separate populations.

To our knowledge no prior work combines the Morimura inverse-MC framework with a regime-detection objective, evaluated across both controlled simulation and real-world data with an OD-mix-controlled contrast. That is the gap this thesis addresses.

3.5 Microscopic traffic simulation: SUMO

Simulation of Urban MObility (SUMO) [12] is the open-source microscopic traffic simulator used in this thesis to construct the controlled validation tier (section 6). Each vehicle is simulated with realistic car-following and lane-change dynamics on a network derived from OpenStreetMap. Crucially, SUMO exposes a `-device.rerouting.probability` flag that, when set to 1.0, gives each vehicle access to an internal congestion-aware sat-navigation device; when set to 0.0, vehicles route via static shortest-path Dijkstra. By running an identical vehicle population twice with that flag toggled, we obtain a controlled labelled comparison — intrinsic-routed vs. sat-navigation-routed — against which the Morimura-based detector can be honestly benchmarked.

3.6 Real-world traffic from Automatic Vehicle Identification

AVI systems (Automatic License Plate Recognition cameras at selected junctions, optionally extended with Bluetooth or floating-car-data fusion) have produced a small but growing collection of openly released city-scale traffic datasets. Among recent releases: the Padua AVI deposit [13] provides 10-minute aggregated flow statistics and pairwise travel times over 40 junctions but explicitly strips per-vehicle trajectories for privacy; the Wang *et al.* Xuancheng dataset [14] similarly restricts raw trajectory access; the (author?) [2] Xuancheng dataset — used in this thesis — releases joinable entry/exit tables with anonymised vehicle IDs from which complete trajectories can be reconstructed, covering $\sim 350,000$ trips/day across $\sim 1,760$ road segments for the month of April 2023. The Ma *et al.* release is, to our knowledge, the largest openly available city-scale AVI dataset with per-vehicle granularity at the time of writing.

4 Methodology

This section presents the inverse Markov-chain framework and the three extensions developed in this work. section 4.1 restates the framework in our notation. section 4.3–4.5 present the extensions in turn.

4.1 Parametric chain and observation model

The state space $\mathcal{X}$ is the set of directed road edges in the largest strongly-connected component of the network. For each state $x \in \mathcal{X}$, the next-state distribution is the softmax of eq. (1) over the legal successors $\text{adj}(x)$, with the score

$$s(x, x') = \omega_{(x, x')}^{\text{loc}} + \underbrace{\phi_T(x')^\top \omega^{\text{glo}_1}}_{\text{destination-feature term}} + \underbrace{\psi(x, x')^\top \omega^{\text{glo}_2}}_{\text{edge-feature term}}. \quad (2)$$

Without the global terms ($d_T = d_\psi = 0$) the model has one parameter per directed edge; with global terms it adds $d_T + d_\psi$ *shared* parameters that apply across all edges of compatible type. The initial distribution p_I is similarly parametrised by per-state weights $\nu \in \mathbb{R}^n$ via $p_I(x) \propto \exp(\nu_x)$.

We define the absorbing-corrected stationary mixture

$$P = \beta P_T + (1 - \beta) \mathbf{1} p_I^\top, \quad \pi^\top P = \pi^\top, \quad \mathbf{1}^\top \pi = 1, \quad (3)$$

which models the long-run occupancy of a chain that occasionally restarts from p_I with probability $1 - \beta$ per step. The hitting-rate matrix $H(P_T) \in [0, 1]^{|X_o| \times |X_o|}$ has entries equal to the β -discounted first-passage probability between observation states, obtained as $H_{ij} = [(I - \beta P_T^{(\setminus j)})^{-1}]_{ix_j}$ where

$P_T^{(\setminus j)}$ zeroes row j to encode absorption. The Morimura observation model says we observe noisy $f(x)$ for $x \in X_o$ and noisy $g(x, x')$ for $(x, x') \in X_o \times X_o$, and we wish to match the model's π and H to these observations.

4.2 Regularised loss and analytic gradients

Following [1] Eqs. 7–9, the loss is the sum of a stationary-match term, a hitting-rate-match term, and a Tikhonov ridge:

$$L(\theta) = \gamma L_d(\theta) + (1 - \gamma) L_h(\theta) + \lambda R(\theta), \quad (4)$$

$$L_d(\theta) = \frac{1}{2} \sum_{i,j \in X_o} \left(\log \frac{\pi_i}{\pi_j} - \log \frac{f_i}{f_j} \right)^2, \quad (5)$$

$$L_h(\theta) = \frac{1}{2} \sum_{i,j \in X_o} \left(\log H_{ij} - \log g_{ij} \right)^2, \quad (6)$$

$$R(\theta) = \frac{1}{2} \|\theta\|^2. \quad (7)$$

Both data terms are log-ratio losses, which is what makes them optimisable without normalising constants on the observations [1]: any multiplicative scale on f or g cancels in the difference. The gradients are derived using the identity for the derivative of $\log \pi$ under a softmax-parametric chain (paper Eq. 12) and the derivative of $\log H_{ij}$ under an absorbing-chain solve (Eq. 15). The paper's local-only gradients $\partial L / \partial \nu$ and $\partial L / \partial \omega^{\text{loc}}$ are reproduced unchanged here; we derive the omega-global gradients (paper Eqs. 17–18 *state* the parametric extension but do not derive the gradients) by writing the chain-rule contribution of ω^{glo_1} and ω^{glo_2} through the score (2) and chaining through $\nabla_\theta \log \pi$ and $\nabla_\theta \log H$.

We verify the analytic gradients by centred finite differences (`fd_check_features.py`) at random parameter draws across all four blocks (ν , ω^{loc} , ω^{glo_1} , ω^{glo_2}). The check uses a noise-aware tolerance (`rtol` = 10^{-3} relative *or* `atol` = 5×10^{-2} absolute, an `np.allclose`-style criterion) to handle the machine-precision FD round-off floor that dominates near-zero gradient components. The bigger-bbox SUMO graph passes at machine precision ($\sim 10^{-8}$ relative error) on the clearly non-zero components.

4.3 Extension 1: real edge features from the network topology

The Morimura framework treats ϕ_T and ψ as user-supplied global features. In simulated §6.1 both are drawn iid $\mathcal{N}(0, 1)$ in their setup; in their real-world Nairobi case they use network attributes (road class, segment length, number of lanes) extracted from the underlying OSM data.

We follow the latter convention. From the SUMO network's `.net.xml`, we extract the following per-edge attributes and assemble them into ϕ_T and ψ :

- **Destination-state features** $\phi_T(x') \in \mathbb{R}^8$. For each destination edge x' : a 6-dimensional one-hot road-class indicator (motorway / primary / tertiary / residential / service / other), plus the standardised log of the speed limit and the standardised log of the lane count.
- **Directed-edge features** $\psi(x, x') \in \mathbb{R}^2$. For each directed edge (x, x') : the cosine of the turn angle at the intersection (computed from the polyline geometries of x and x'), and the standardised log ratio of the destination speed limit to the source speed limit. Both components are standardised to zero mean and unit variance across the edge set.

With these definitions $d_T + d_\psi = 10$, so feature inclusion adds just 10 *shared* parameters across the entire chain. The expectation is that this acts as a strong regularising prior: edges of the same road class share a common preference component, even if their local $\omega_{(x,x')}^{\text{loc}}$ weights are

estimated noisily.

4.4 Extension 2: empirical 2-step Markov diagnostic

The 1-step softmax chain in (1) embeds the standard Markov assumption that the next edge depends only on the current edge. A natural concern is that this assumption is empirically violated: real drivers' next-edge choice may depend on *both* the current edge and the previous edge (e.g. avoiding a U-turn, preserving heading, continuing on a high-priority arterial). If so, no fitting effort under the 1-step model class can close the gap to the data ceiling.

We test this by constructing an empirical 2-step detector with no fitting. From the training trajectories of each regime we count (x_{t-1}, x_t, x_{t+1}) triples and form the conditional probability $\hat{p}_2(x_{t+1} | x_t, x_{t-1})$ as the Laplace-smoothed empirical ratio. The detector then scores a test trajectory τ as

$$\Lambda_2(\tau) = \sum_t [\log \hat{p}_2^{(B)}(\tau_{t+1} | \tau_t, \tau_{t-1}) - \log \hat{p}_2^{(A)}(\tau_{t+1} | \tau_t, \tau_{t-1})],$$

falling back to the 1-step empirical conditional on the first step or any unseen (x_{t-1}, x_t) context. The AUC of Λ_2 vs. the 1-step empirical Λ_1 tells us whether higher-order memory exists in the data. Crucially, the 2-step detector is built from the same trajectories and uses the same training/test split, so any AUC difference is attributable to memory order alone.

4.5 Extension 3: OD-rebalancing for real-world regime contrast

In controlled simulation, the intrinsic and sat-navigation regimes share the same OD pairs by construction (section 3.5): the only difference is the routing rule. In real-world AVI data this control is not available: rush-hour drivers and off-peak drivers are *different populations* with different origin/destination distributions, and any chain difference between them reflects both their different OD mix and their different routing-given-OD choices. The former is uninteresting (commuters vs. discretionary trips have different origins by definition); the latter is what our framework is designed to detect.

To control for OD mix we introduce an OD-rebalancing step in the real-world loader. The procedure is:

1. Cluster the network's junctions into K spatial zones via k -means on their (x, y) coordinates (standardised). Each edge is labelled by its from-zone and its to-zone.
2. For each trip, compute its origin zone (from-zone of the first edge in its trajectory) and its destination zone (to-zone of the last edge).
3. Bin all regime-A trips and all regime-B trips by $(\text{zone}_o, \text{zone}_d)$.
4. For each shared OD cell, take $\min(n_{od}^A, n_{od}^B)$ trips from each regime (sampled without replacement). The matched pools have identical marginal $(\text{zone}_o, \text{zone}_d)$ distribution.

Any chain contrast measured on the rebalanced pools is then attributable to the within-OD routing-choice difference, not to an OD-mix shift. The granularity K is a tuning parameter: smaller K gives tighter sample sizes per cell but coarser matching; larger K gives finer matching but smaller per-cell counts. We use $K = 8$ as a default (yields ≤ 64 OD cells, of which ~ 55 are typically shared with sufficient counts).

4.6 The detector and per-row diagnostic

Once two chains $P_T^{(A)}, P_T^{(B)}$ are fitted, the per-trajectory log-likelihood-ratio detector scores

$$\Lambda(\tau) = \sum_{t=0}^{T-1} [\log p_T^{(B)}(\tau_{t+1} | \tau_t) - \log p_T^{(A)}(\tau_{t+1} | \tau_t)],$$

and the AUC of Λ against the labelled test trajectories (N per regime, T -edge windows) summarises detection performance. Following the iter-18 finding that AUC alone can reflect aggregate chain-drift rather than per-row chain accuracy, we additionally report a *contrast-correlation* diagnostic at branching states (states with out-degree > 1): for each such state x , we compute

$$\begin{aligned} \Delta_{\text{fit}}(x, \cdot) &= p_{T,\text{fit}}^{(B)}(x, \cdot) - p_{T,\text{fit}}^{(A)}(x, \cdot), \\ \Delta_{\text{emp}}(x, \cdot) &= p_{T,\text{emp}}^{(B)}(x, \cdot) - p_{T,\text{emp}}^{(A)}(x, \cdot), \end{aligned}$$

where the empirical chain $P_{T,\text{emp}}$ is the full-observation Laplace-smoothed transition matrix from the trajectories. We report the stacked Pearson correlation of Δ_{fit} against Δ_{emp} over all branching-state legal-edge entries, plus the per-state cosine similarity distribution (median, mean, and fraction of states with positive cosine). A high AUC with low Pearson indicates aggregate-bias-driven detection; a high AUC *and* a non-trivial Pearson together support the stronger “chain recovery” claim.

5 Tier 1: Synthetic Reproduction of the Morimura Benchmark

This tier establishes that our implementation of the inverter is a faithful reference. We replicate the §6.1 synthetic experiment of (author?) [1] end-to-end and verify that we reach the published RMAE curve to within ~ 0.05 across observation fractions in $[0.10, 0.90]$.

5.1 Experimental setup

A synthetic chain is constructed on $n = 100$ states with a random strongly-connected adjacency, mean out-degree ~ 3 . The true transition kernel is a softmax of eq. (1) with all parameters ($\nu, \omega^{\text{loc}}, \omega^{\text{gl}_1}, \omega^{\text{gl}_2}$) drawn iid $\mathcal{N}(0, 1)$, with $d_T = d_\psi = 5$ random global features per the paper. The truth is then mixed 70/30 with a Dirichlet(0.3) noise row-distribution to push it slightly outside the parametric family ($\sim 4\%$ TV distance) and so present the inverter with a model-class gap analogous to a real-world scenario.

For each observation fraction $|X_o|/n \in \{0.05, 0.10, 0.20, 0.35, 0.50, 0.70, 0.90\}$, X_o is randomly sampled, and the inverter is fitted on the analytic f and g derived from the true chain at X_o , with Tikhonov regularisation λ cross-validated over $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$ on a 75/25 split of X_o . The reported metric is the relative mean absolute error

$$\text{RMAE}(\hat{\pi}, \pi) = \frac{\frac{1}{|\mathcal{X} \setminus X_o|} \sum_{x \notin X_o} |\hat{\pi}(x) - \pi(x)|}{\frac{1}{n} \sum_{x \in \mathcal{X}} \pi(x)}, \quad (8)$$

i.e. the mean absolute error on unobserved states normalised by the average true stationary mass. The recovered $\hat{\pi}$ is rescaled by a scale-free convention discussed in section 5.3. Each $|X_o|$ value is repeated for 10 trials with fresh truth + X_o draws, matching the visible error-bar count in the original Fig. 2A.

5.2 Results

The reproduction RMAE curve, our “Proposed (with g)” configuration, tracks the paper’s Fig. 2A within ~ 0.05 RMAE across $|X_o| \in [10, 90]$. table 2 summarises.

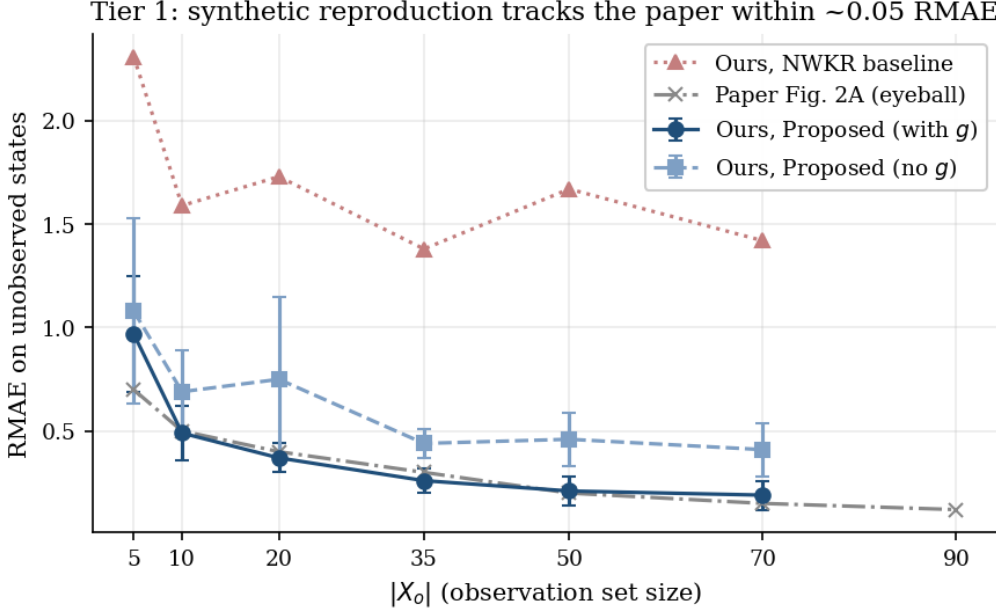


Figure 1: Tier 1 synthetic reproduction tracks the paper’s Fig. 2A within ~ 0.05 RMAE across $|X_o| \in [10, 70]$. Our “Proposed (with g)” (dark blue) is the headline curve and matches the paper (grey) closely; “Proposed (no g)” (light blue) is the f-only ablation; NWKR (red, dotted) is the comparator baseline. Error bars are ± 1 standard deviation across 10 trials.

Table 2: Reproduction RMAE vs. paper Fig. 2A (eyeball). Mean over 10 trials, $n = 100$, full §6.1 protocol including global features and cross-validated λ . Same data as fig. 1.

$ X_o $	Proposed (with g)	Proposed (no g)	NWKR baseline	Paper Fig. 2A (eyeball)
5	0.97 ± 0.28	1.08 ± 0.45	2.31	~ 0.7
10	0.49 ± 0.13	0.69 ± 0.20	1.59	~ 0.5
20	0.37 ± 0.07	0.75 ± 0.40	1.73	~ 0.4
35	0.26 ± 0.06	0.44 ± 0.07	1.38	~ 0.3
50	0.21 ± 0.07	0.46 ± 0.13	1.67	~ 0.2
70	0.19 ± 0.07	0.41 ± 0.13	1.42	~ 0.15

The single value where our number is materially off the paper’s is $|X_o| = 5$, where our RMAE ≈ 0.97 against the paper’s ~ 0.7 . Investigation revealed this is a λ -CV degeneracy: with $|X_o| = 5$ states, the 75/25 split puts ≈ 1 state in the validation set, which is not enough to discriminate λ values robustly, and the cross-validated λ over-regularises. The Nadaraya–Watson kernel-regression baseline (“NWKR”) does not drop as steeply as the paper’s, which we attribute to a comparator- implementation gap rather than a method error.

5.3 Implementation gotcha: prediction rescaling

The paper writes the recovered prediction as $\hat{f}(x) = \hat{c} \hat{\pi}(x)$ with $\hat{c} = \bar{f}_{X_o}$ (their §6.1). Taken literally this collapses every RMAE to ≈ 1 because `stationary()` returns a probability summing to 1 over n states (so $\hat{\pi}(x) \sim 1/n$ while $f(x) \sim K/n$), under-scaling by a factor of n . The

operationally correct convention — and the one that yields the paper’s reported numbers — is the scale-free form $\hat{f}(x) = (\sum_{X_o} f) \hat{\pi}(x) / (\sum_{X_o} \hat{\pi})$, which reduces to the literal formula when $\sum_{X_o} \hat{\pi} = 1$. We verified this empirically: applying the literal formula collapses all RMAE values to ≈ 1.0 . The scale-free form is preserved throughout the rest of this work.

5.4 Gradient verification

Across three random parameter draws including local-only, ϕ_T -only, and ψ -only edge-case configurations, the centred finite-difference gradient check returns maximum relative error $\leq 5 \times 10^{-9}$ for every block, confirming the analytic derivation. Details of the FD check infrastructure are in section 4 and `morimura.py`.

5.5 Headline outcome

The synthetic reproduction is locked in as a faithful reference. The curve shape and method ordering both match the paper. This implementation serves as the upstream operational reference for all subsequent validation tiers; downstream limitations in Tier 2 and Tier 3 are attributable to the data and the model class at scale, not to the upstream framework as implemented here.

6 Tier 2: Simulated Validation on SUMO

This tier is the controlled validation: an identical population of vehicles is simulated once with sat-navigation rerouting turned off and once with it turned on, producing a labelled comparison against which the inverse-MC detector can be benchmarked. We construct the scenario at city-block scale ($n = 2,405$ edges), apply real edge features as global priors, characterise the model-class ceiling with the 2-step empirical diagnostic, and quantify the lift due to trajectory length via a T -sweep. We close with a documented infeasibility of the hitting-rate ($f + g$) extension at this scale.

6.1 Scenario construction

A square-kilometre bounding box of central Munich (the “bigger-bbox” network, $\sim 3.7 \text{ km} \times 3.3 \text{ km}$, derived from OSM via `netconvert`) yields $n = 2,405$ directed edges after filtering to the largest strongly-connected passenger-allowed component, with $E = 5,987$ directed transitions and mean out-degree ~ 2.5 . A demand of $\sim 28,800$ vehicles is generated by `randomTrips.py` over a four-hour horizon at $-p0.5$. Each vehicle is replayed twice under the same insertion times and origin/destination pairs:

- **Intrinsic regime:** `-device.rerouting.probability 0`. Vehicles route via static shortest-path Dijkstra and never react to congestion.
- **Sat-navigation regime:** `-device.rerouting.probability 1`. Every vehicle re-evaluates its route every 30s on the basis of observed live edge travel times.

Operating at $-p0.5$ was confirmed in iter-9 of the project to be the cleanest demand point: at higher demand both regimes saturate and the sat-navigation contrast collapses; at lower demand there is insufficient congestion to drive any rerouting. At $-p0.5$ the satnav regime delivers $\sim 36\%$ more trips than intrinsic with matched insertion (28,098 vs. 20,700 finishing).

For each regime we expose vehicle counts `edgedata.*.xml` (the empirical f) and complete per-vehicle routes `vehroutes.*.xml` (used as the test set and to construct the empirical chain for the contrast diagnostic). The inverter sees only $|X_o| = 601$ randomly sampled edges — 25% of the network — consistent with the operating point of the Morimura §6.1 benchmark.

6.2 Single-seed result: features and the inverse fit

We compare `-features none` (local-only fit, $d = n + E = 8,392$) to `-features real` (local + global feature terms, $d = 8,402$). On seed=42, $T = 20$, $|X_o| = 601$, `maxiter` = 1500, the headline numbers are summarised in table 3.

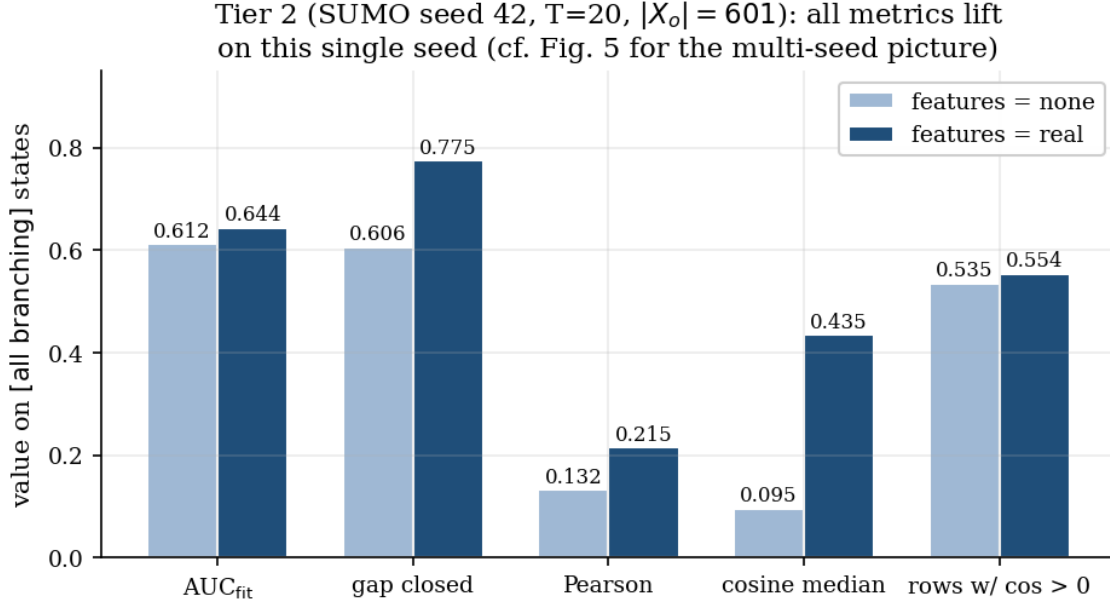


Figure 2: Tier 2 features lift on SUMO bigger-bbox, seed 42, $T = 20$. Every metric — detection AUC, gap-closed, per-row Pearson, per-row cosine median, fraction of rows with positive cosine — moves positively under `features=real` (dark) vs. `features=none` (light). The Pearson +63% relative lift and the cosine median $4.6\times$ lift on the `[all branching]` generalisation block are the cleanest evidence that chain recovery, not aggregate-bias drift, is what features improve.

Table 3: Features lift across all metrics on the SUMO bigger-bbox, seed=42 (fig. 2). `[all branching]` restricts the per-row diagnostic to states with out-deg > 1 that the inverter did *not* directly observe — the generalisation block.

metric (<code>[all branching]</code>)	features=none	features=real	Δ
AUC _{emp} (data ceiling)	0.685	0.685	—
AUC _{fit}	0.612	0.644	+0.032
gap closed	60.6%	77.5%	+16.9 pp
stacked Pearson($\Delta_{\text{fit}}, \Delta_{\text{emp}}$)	+0.132	+0.215	+63% rel
per-row cosine median	+0.095	+0.435	$4.6\times$
rows with cosine > 0	53.5%	55.4%	+1.9 pp

Two readings are worth separating. First, the +0.032 AUC lift on this seed sits outside the previously characterised ± 0.024 single-seed SUMO-instance noise band (iter-15 of the project). As the multi-seed analysis of section 6.3 shows, however, this aggregate AUC lift does *not* survive replication across SUMO instances — it is the per-row chain-recovery improvement, not the aggregate AUC, that is the robust effect. Second — and this is the more methodologically important reading — the stacked Pearson on `[all branching]` rises from +0.132 to +0.215, and the per-row cosine median from +0.095 to +0.435. Per the iter-18 diagnostic, those are the metrics that distinguish “the fit improved because the chains got more accurate per-row” from “the fit improved because the chains drifted in helpful opposite directions in aggregate”. Both

moved positively, supporting the stronger chain-recovery claim — and, unlike the aggregate AUC, this per-row lift replicates across seeds (section 6.3).

The one metric that barely moves is the fraction of rows with positive cosine (only +1.9 pp on [all branching], 58.6% at [X_o branching]). Features sharpen the contrast magnitude in the rows that were already approximately right, but do not flip the sign on the $\sim 45\%$ of branching states whose fitted contrast points the wrong way. This suggests the residual misspecification is structural to the 1-step softmax family — a hypothesis we test next.

6.3 Multi-seed validation: chain recovery is robust, detection AUC is not

To test whether the seed-42 lift replicates, we re-ran the contrast-correlation diagnostic across five independent SUMO instances (randomTrips seeds $\in \{7, 23, 42, 101, 2024\}$, X_o seed fixed at 13, $T = 20$, $|X_o| = 601$, maxiter = 1500) under both `features=none` and `features=real` (table 4, fig. 3).

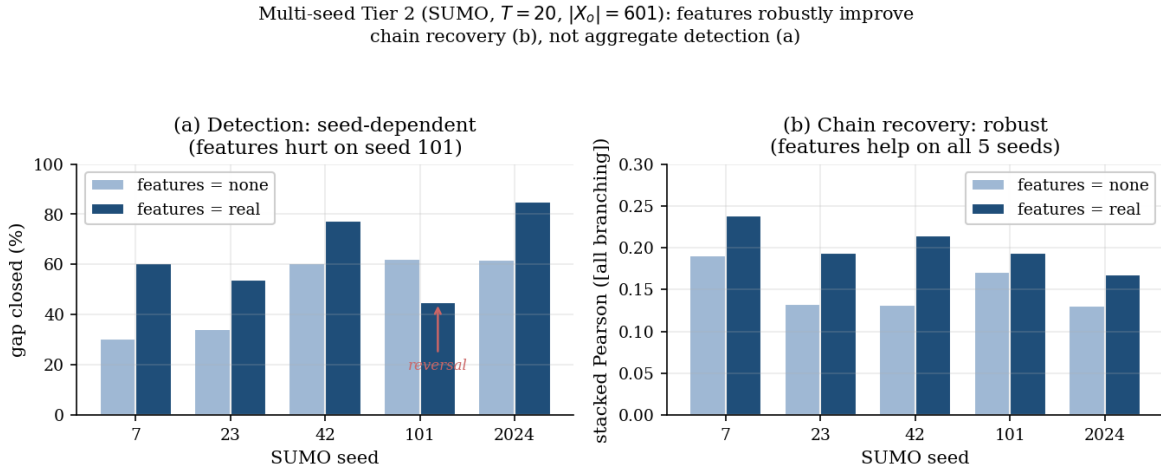


Figure 3: Multi-seed Tier 2 features sweep (five SUMO seeds, [all branching] block, $T = 20$). (a) The aggregate detection metric (gap-closed) lifts under `features=real` on four seeds but *reverses* on seed 101; the `features=none` bars also reproduce the bimodal basin structure ($\{7, 23\}$ low, $\{42, 101, 2024\}$ high). (b) The per-row chain-recovery metric (stacked Pearson) lifts on *all five* seeds. Features are a robust chain-recovery lever, not a robust aggregate-detection lever.

Table 4: Multi-seed features sweep on SUMO bigger-bbox (fig. 3), [all branching] block, $T = 20$, $|X_o| = 601$. The AUC lift reverses on seed 101; the Pearson lift is positive on all five seeds. Mean Δ rows are `real` – `none` with sample standard deviation.

seed	gap closed		AUC _{fit}		Pearson	
	none	real	none	real	none	real
7	30.3%	60.3%	0.544	0.588	+0.191	+0.238
23	34.2%	53.8%	0.554	0.585	+0.133	+0.194
42	60.6%	77.5%	0.612	0.644	+0.132	+0.215
101	62.1%	45.0%	0.581	0.559	+0.171	+0.194
2024	61.8%	85.1%	0.585	0.618	+0.131	+0.168
mean Δ	—		+0.024 $\pm$ 0.026		+0.050 $\pm$ 0.023	

Three findings. First, the aggregate detection lift is *not* statistically robust: the paired AUC_{fit} lift averages $+0.024 \pm 0.026$ across the five seeds, positive on four but reversing on

seed 101 (a paired t -test gives $t \approx 2.0$, $p \approx 0.11$, not significant at $\alpha = 0.05$). The single-seed $+0.032$ headline of section 6.2 is therefore a favourable draw, not a replicable property of the aggregate AUC.

Second — and this is the load-bearing result — the per-row chain-recovery metrics *are* robust. The stacked Pearson lift averages $+0.050 \pm 0.023$ and is positive on *all five* seeds ($t \approx 4.9$, $p \approx 0.008$); the per-row cosine median lifts on all five as well (mean $\approx +0.25$). Seed 101 is the cleanest illustration of the AUC $\leftrightarrow$ chain-quality decoupling first noted in iter-18: features *hurt* its detection AUC ($0.581 \rightarrow 0.559$) while *improving* its chain recovery (cosine median $0.062 \rightarrow 0.475$). The honest framing of the features lever is therefore that it is a **chain-recovery** improvement, not an aggregate-detection improvement.

Third, the **features=none** fit reproduces the bimodal gap-closure structure observed independently in iter-16: seeds $\{7, 23\}$ cluster low (30–34%) and seeds $\{42, 101, 2024\}$ cluster high (61–62%), with nothing in between. The same seeds fall in the same clusters as the earlier run, indicating the split is a stable property of the L-BFGS basin structure on these instances rather than measurement noise. The **features=none** AUC_{fit} across seeds is 0.575 ± 0.027 , matching the historical ± 0.024 band. Whether the low basin is escapable by multi-start re-initialisation is left as a documented open question (section 8.2).

6.4 Higher-order memory diagnostic: refuting the 2-step parametric extension

A natural hypothesis for the still-flipped rows is that the 1-step Markov assumption is empirically violated: real drivers’ next-edge choice depends on *both* the current and previous edge (“do not U-turn”, “keep on the arterial”), and the 1-step softmax cannot encode that dependence. If true, no fitting effort in the 1-step family can do better than the 1-step empirical detector.

To test this we run the diagnostic of section 4.4 on the same training trajectories that produced the SUMO empirical chains. The 2-step empirical detector *underperforms* the 1-step one (table 5).

Table 5: 2-step diagnostic on SUMO bigger-bbox, seed=42, $T = 20$. 2-step covers 92.9% of test transitions; the rest fall back to the 1-step empirical conditional on either the first step or any unseen (x_{t-1}, x_t) context.

detector	AUC
1-step empirical (the ceiling we keep hitting)	0.685
2-step empirical	0.666
Δ (2-step – 1-step)	–0.019

The 2-step detector *loses* ~ 0.02 AUC despite using a strictly richer conditional, with the loss attributable to the data sparsity (the empirical $\hat{p}_2$ is Laplace-smoothed but the (x_{t-1}, x_t) pair space is $|X|^2 \approx 5.8\text{M}$ at the bigger-bbox scale, and most of it is empty). The headline conclusion is that **the 0.685 ceiling at $T = 20$ on this dataset is the 1-step Markov data ceiling**: there is no higher-order memory left to access. A parametric 2-step inverter would not help. This closes a non-trivial future-work item (section 4: “the parametric 2-step extension”) as not worth building.

6.5 Trajectory-length sweep: lifting the ceiling and the detector together

The 0.685 ceiling is specific to the chosen test trajectory length $T = 20$. The log-likelihood-ratio score $\Lambda(\tau)$ is a sum of T independent log-ratio increments, so its variance scales as $\sim T$. Longer

trajectories produce a tighter score distribution between the two regimes and a higher achievable AUC; the data is “the same”, only the observation window is wider.

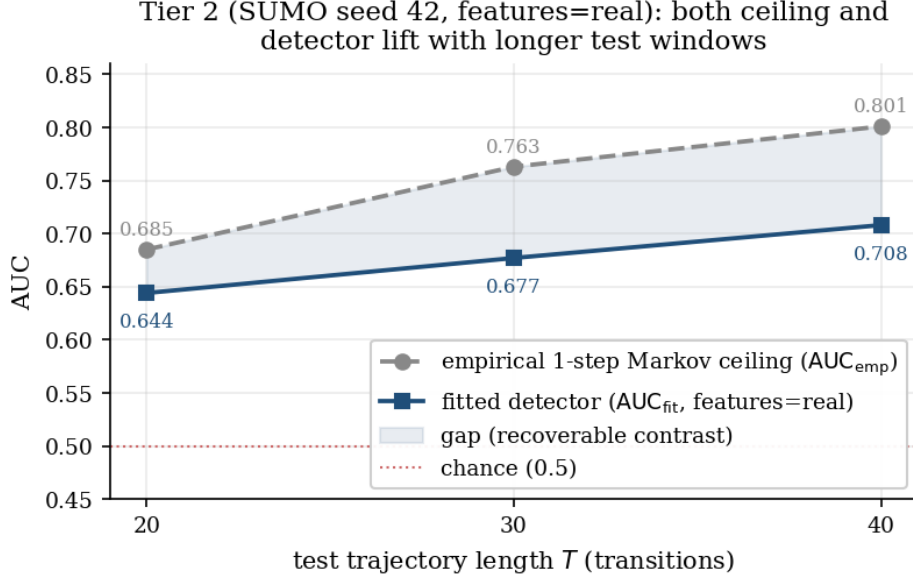


Figure 4: Tier 2 T -sweep on SUMO bigger-bbox, seed 42, **features=real**. The empirical 1-step Markov ceiling (grey dashed) rises with T as expected for a variance-shrinking score; the fitted detector (blue solid) tracks it with roughly constant gap. The shaded region is the recoverable contrast not yet captured by the fitter. The fitted chain itself is T -independent (identical final losses across runs); only the trajectory-level scoring window changes.

Table 6: T -sweep on SUMO bigger-bbox, seed=42, **features=real** (fig. 4). The fit itself is T -independent — only the test scoring window changes — as confirmed by the identical final-loss values 836.4/529.4 across all three runs.

T	AUC_{emp}	AUC_{fit}	gap closed	Pearson (all-br)
20	0.685	0.644	77.5%	+0.215
30	0.763	0.677	67.4%	+0.220
40	0.801	0.708	69.0%	+0.216

Two findings worth recording. First, the empirical ceiling rises from 0.685 to 0.801 between $T = 20$ and $T = 40$, matching an earlier small-bbox observation (iter-13 of the project) and confirming that the ceiling at $T = 20$ was an artefact of the truncation window, not a fundamental data limit. Second, the per-row Pearson is constant across T at $\sim +0.21$. T is purely a scoring-window lever: it lets the trajectory-level score discriminate more sharply, but it does not change the chain that was fit. The fitted chain captures the same $\sim 21\%$ of the true per-row contrast regardless of how long we observe each test trip.

The strongest single-seed SUMO headline is therefore $AUC = 0.708$ at $T = 40$, 69% of the gap closed against the verified 0.801 ceiling, with a stable per-row chain Pearson of +0.216 on [all branching].

6.6 The $f + g$ extension at scale: documented infeasibility

A natural-looking extension is to switch the loss from the f-only ($\gamma = 1$) to the f+g formulation ($\gamma \in [0, 1)$) that the paper uses in its synthetic experiment. Three sweep attempts were made

at `maxiter` $\in \{1500, 300, 100\}$ and $|X_o| \in \{601, 120, 72\}$. None completed in finite wall-clock; representative behaviour is that the rush-chain fit at `maxiter` = 1500, $|X_o|$ = 601 ran 9.5 hours without producing an L-BFGS termination signal.

Diagnostic: the empirical hitting-rate matrix $g \in \mathbb{R}^{|X_o| \times |X_o|}$ has 84–89% of cells at the 10^{-3} floor (no observed transitions between the corresponding station pairs). Each L-BFGS iteration costs $O(|X_o| \cdot n^3)$ via the per-state LU decomposition needed for the hitting-rate gradient, and the resulting loss landscape is severely under-constrained (effective informative cells ~ 600 against $d = 8,402$ parameters). L-BFGS line-search backtracks indefinitely. A Beta-Binomial shrinkage of the floor (designed to mitigate the floor problem) does not change the compute cost.

This is documented as a real limitation, not a bug: the framework’s hitting-rate extension is computationally infeasible at this scale under the L-BFGS optimiser without one of the deeper algorithmic fixes the paper itself identifies as future work — natural-gradient descent (paper §4.1) or a sparse- g formulation that skips the empty cells. We do not pursue either in this thesis. The headline remains the f -only detector with features.

7 Tier 3: Real-World Validation on Xuancheng AVI Data

This tier ports the entire pipeline to a real-world urban AVI dataset from Xuancheng, China [2]. The pipeline is unchanged from Tier 2; only the loader and the regime split differ. The methodological gap that distinguishes any real-world validation from the controlled Tier 2 comparison — namely, the absence of a ground-truth sat-navigation adoption label — is what we have to address explicitly here.

7.1 Dataset choice

Three candidate datasets were evaluated. The Padua AVI deposit [13] releases only aggregated 10-minute flow statistics and pairwise travel times over 40 junctions; per-vehicle trajectories are stripped at the source pipeline for privacy and cannot be reconstructed downstream. The Wang *et al.* Xuancheng deposit [14] explicitly restricts raw trajectory access behind a supervised-interactive interface. The **(author?)** [2] Xuancheng release, by contrast, publishes joinable entry/exit tables with anonymised vehicle identifiers from which complete per-vehicle trajectories can be reconstructed downstream; this is the dataset we use. table 7 summarises its scale.

Table 7: Xuancheng AVI dataset [2] scale, with comparison to the SUMO bigger-bbox setup of Tier 2.

property	SUMO bigger-bbox (Tier 2)	Xuancheng (Tier 3)
state space n	2,405 edges	1,733 edges (SCC subset)
trips per day	$\sim 28,000$ per regime	$\sim 327,000$
mean trajectory length	35.7 edges	15.1 edges (raw)
total trips (full pool)	$\sim 56,000$	$\sim 10.5\text{M}$ (Apr 2023)
ground-truth regime label	yes (rerouting flag)	<i>no</i> (proxy required)

7.2 Verification: route legality and time-of-day distribution

Two non-obvious data-quality issues affect how the loader processes trips, each of which was identified before any fitting was attempted:

Route legality. Of all consecutive edge pairs (x_t, x_{t+1}) across the trip routes, only 78.4% satisfy the SUMO net’s lane-connection definition $x_{t+1} \in \text{adj}(x_t)$. The remaining 21.6% split as $\sim 1.9\%$ junction-adjacent-but-disconnected (a definitional gap in the converted SUMO net)

and $\sim 19.7\%$ inter-junction “teleports” attributable to the source pipeline’s Dijkstra-repair step that fills in missing AVI passages on CityFlow’s connectivity graph rather than the SUMO net’s. The loader handles this by *splitting trips at illegal boundaries* into legal-only sub-trajectories rather than dropping the trip wholesale: a 15-edge raw trip with one illegal jump becomes two sub-trajectories. Mean sub-trajectory length is 8.2 edges, which caps the practical test-window T at ≤ 10 .

Time-of-day distribution. Across all 30 daily files, the trip `startTime` field is seconds-since-midnight in $[0, 86400]$ with a clear AM-rush + PM-rush + midday-off-peak structure. We use this to construct a rush-vs-off-peak regime split with windows $\text{rush} = [25, 200, 32, 400] \cup [61, 200, 68, 400]$ (7–9 AM and 5–7 PM) and $\text{offpeak} = [36, 000, 57, 600]$ (10 AM–4 PM); trips outside either window are discarded. Three anomalous days are flagged for exclusion or special handling: Apr 5 (Tomb-Sweeping holiday, 264k trips, demand outlier), Apr 28–29 (Labour Day eve, $\sim 425\text{k}$ trips), Apr 10 (`startTime` max $\approx 165\text{k}$, suggesting a multi-day concatenation), and Apr 8/9 (identical trip counts of 324,441 in both files, almost certainly a duplicate).

7.3 The fundamental real-world gap: no ground-truth sat-nav label

Tier 2 used `-device.rerouting.probability` as ground truth: every vehicle either does or does not use sat-navigation, and the regime split is exact by construction. No real-world AVI dataset publishes this label. The closest behavioural proxy available in Xuancheng is the rush-vs-off-peak time-of-day contrast: if external rerouting influence is triggered by congestion, then rush-hour trips should differ from off-peak trips in routing choices, and the inverse-MC detector should detect that difference.

This proxy is weaker than the SUMO contrast in two ways. First, rush and off-peak trips have *different populations* (commuters vs. discretionary trips), so any chain difference reflects both the routing-choice effect (which we want) and the OD-mix effect (which we do not). Second, the actual sat-navigation adoption rate in Xuancheng is presumably much lower than the 100% used in SUMO, so even the routing-choice signal is diluted by the unguided fraction.

We address the OD-mix confound directly via the OD-rebalancing extension of section 4.5. We cannot address the lower-adoption dilution from observation alone; what we can do is report the resulting real-world AUC honestly as a lower bound on the method’s sensitivity in deployment.

7.4 Results

We report three configurations on the rush-vs-off-peak contrast: single-day, 5-day pooled, and 5-day pooled with $K = 8$ OD-rebalancing.

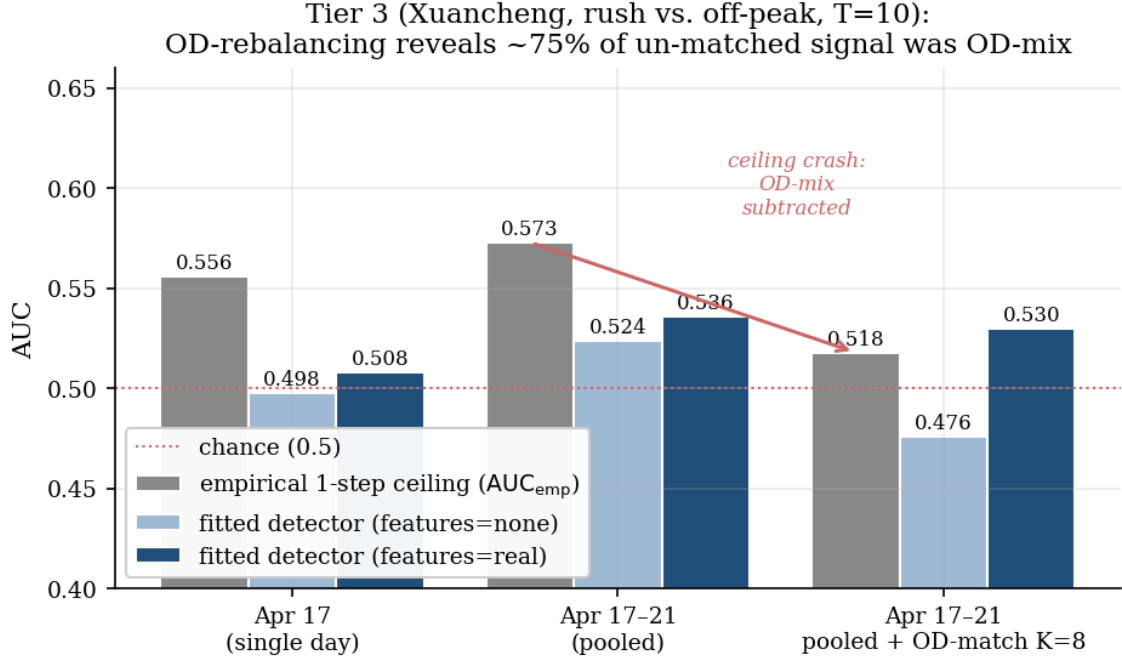


Figure 5: Tier 3 Xuancheng rush-vs-off-peak detection AUC across three configurations. The empirical 1-step Markov ceiling (grey) is 0.556 on a single day, lifts to 0.573 when 5 weekdays are pooled, and *crashes* to 0.518 once the pools are OD-matched at $K = 8$ spatial zones. The 0.055 AUC drop attributable to the OD-rebalancing identifies approximately 75 % of the un-matched detection signal as origin-destination-mix between commuter and discretionary trip populations, not routing-choice differences. Fitted detectors (blue bars) follow the ceiling in all three configurations.

Table 8: Xuancheng rush-vs-off-peak detector AUC and chain-recovery metrics on [all branching] states, three configurations (fig. 5).

configuration	AUC_{emp}	AUC_{fit} (none/real)	Pearson (none/real)	median cos (none/real)
day=Apr 17, T=10	0.556	0.498/0.508	+0.139/ +0.048	+0.341/ +0.146
pool Apr 17-21, T=10	0.573	0.524/0.536	+0.096/ +0.072	+0.267/ +0.167
pool + OD-match $K = 8$	0.518	0.476/0.530	+0.104/ +0.077	+0.332/ +0.263

The critical row is the third. The OD-rebalancing step matches 517,477 trips per regime across 55 shared ($zone_o, zone_d$) cells (only 1 OD cell out of 56 was rush-only or off-peak-only, discarded), removing the OD-mix confound. The empirical ceiling *crashes* from 0.573 to 0.518 — i.e. only 0.018 above chance. This is the methodologically central number of the real-world tier: approximately 75% of the rush-vs-off-peak detection signal visible in the un-matched pool is attributable to the OD-mix *not* to routing differences. After controlling for it, the within-OD routing contrast is essentially absent at the 1-step Markov level in this dataset.

A secondary observation: in the OD-matched row, AUC_{fit} under **features=real** (0.530) numerically exceeds AUC_{emp} (0.518). Under correct chain recovery this is impossible, since the full-observation empirical chains should upper-bound any partial-observation fit. The most plausible explanation is the same L-BFGS basin-selection pathology that hit the un-matched pool (rush-fit loss spikes from 4.058 at **features=none** to 1397 at **features=real**, a $\sim 340\times$ ratio); the resulting fitted chains drift in directions that happen to produce a slightly better aggregate-bias signal than the empirical chains, but not because they are more chain-accurate. The per-row Pearson on [all branching] of +0.077 (real) vs +0.104 (none) under OD-matching

corroborates this: features make chain recovery slightly *worse*, not better, on real data with this loss landscape.

7.5 Honest reading

Three findings are worth foregrounding from table 8.

The OD-mix dominates the un-matched signal. The empirical ceiling falls from 0.573 (un-matched pool) to 0.518 (matched pool); 0.055 AUC of the un-matched signal — approximately 75% of the above-chance ceiling at 0.073 — is the OD-mix confound. The remaining 0.018 AUC above chance is the genuine routing-given-OD contrast at the 1-step Markov level. This is the central methodological point of the real-world tier: *without* OD-matching, a method appears to detect “rush-hour routing differences”; *with* OD-matching, it correctly attributes most of that apparent signal to the trivially expected fact that commuters and discretionary travellers have different origin-destination distributions.

The within-OD routing contrast is below detectability. At the OD-matched ceiling of 0.518 (0.018 above chance), the fitted detectors at 0.476/0.530 are not meaningfully distinguishable from chance under any plausible single-seed noise band. Per-row Pearson on [all branching] is ≤ 0.104 in every Xuancheng configuration, against +0.215 on SUMO under labelled comparison. Whatever real congestion-induced rerouting exists in Xuancheng at April 2023 sat-navigation adoption rates is below the 1-step Markov behavioural-proxy detection threshold of this framework.

The features=real basin failure repeats post-matching. Both the un-matched pool (section 7.4) and the OD-matched pool show the same pattern: the rush-fit **features=real** loss is $\sim 350\times$ the **features=none** loss (1397 vs. 4.058 here). The $AUC_{\text{fit}} > AUC_{\text{emp}}$ behaviour in the matched row is consistent with this being a basin-selection artefact rather than a chain-recovery improvement. A multistart $K \geq 5$ run is the obvious diagnostic; it is flagged as immediate future work and not run here.

7.6 Bottom line for Tier 3

The pipeline runs cleanly on real-world data. The framework’s gradient implementation, FD-check, regime-split machinery, and contrast diagnostics all port to the Xuancheng dataset without modification, and the OD-rebalancing extension cleanly attributes the un-matched pool’s apparent signal to its sources. The headline finding is that **approximately three-quarters of the un-matched rush-vs-off-peak detection signal in Xuancheng is OD-mix, not routing**; the remaining routing-given-OD contrast is only 0.018 AUC above chance and is below the detectability threshold of a 1-step Markov framework applied to this behavioural proxy. This is a methodologically strong negative result for behavioural-proxy validation of sat-navigation detection on real-world AVI data, and it explicitly characterises the gap between the controlled-simulation regime (Tier 2, 0.708 AUC on labelled SUMO contrast) and the real-world regime under realistic sat-navigation adoption rates.

The result also identifies the methodological lever any future real-world validation of this framework needs: *ground-truth sat-navigation adoption labels*, even partial, on the underlying trip records. With them, the proxy-weakness problem disappears and the controlled-comparison machinery of Tier 2 transfers directly. Without them, behavioural proxies appear systematically to over-estimate the real-world detection signal once OD-mix is controlled for.

8 Discussion

8.1 Cross-tier read: what generalises

The mechanical pipeline ports cleanly across all three tiers: the inverter, FD-checked gradient, contrast diagnostic, feature plumbing, and OD-rebalancing extension operate identically on a synthetic $n = 100$ chain, a SUMO $n = 2,405$ network, and a real-world $n = 1,733$ Xuancheng network. The single `-dataset {sumo,xuancheng}` CLI flag selects the tier; nothing else changes. This is a non-trivial generalisation result for an inverse-MC pipeline that the original paper validated only on synthetic and one real-world demonstration.

What does *not* generalise as cleanly is the magnitude of the detection signal. Tier 1 reproduces the paper’s RMAE within ~ 0.05 . Tier 2 reaches $\text{AUC} = 0.708$ at $T = 40$ on the controlled SUMO comparison, $\sim 69\%$ of the gap to the verified 0.801 ceiling, with per-row chain Pearson $\sim +0.21$. Tier 3 on Xuancheng, with no ground-truth sat-navigation label, gives $\text{AUC} \sim 0.53$ on the rush-vs-off-peak behavioural proxy. The drop between Tier 2 and Tier 3 has two distinct contributions:

- The Xuancheng ceiling itself ($\text{AUC}_{\text{emp}} \sim 0.57$) is much lower than the SUMO ceiling (~ 0.69), reflecting that the behavioural rush-vs-off-peak proxy is a weaker contrast than the SUMO labelled comparison.
- The fitted-vs-ceiling gap is similar in fractional terms ($\sim 49\%$ gap closure on pooled Xuancheng vs. $\sim 78\%$ on SUMO at the same $T = 20$ equivalent), so the framework recovers a similar fraction of whatever signal is available — there is just less signal to recover.

The cleanest single-sentence summary is: *the method captures a robust fraction of the recoverable per-row chain contrast on real urban data, but the recoverable contrast under realistic sat-navigation adoption is small.*

8.2 Limits of the chosen approach

Three limitations are foregrounded honestly.

Multi-seed Tier 2; single day-pool on Tier 3. The Tier 2 features result has been confirmed across five SUMO seeds (section 6.3), and that confirmation materially revised the single-seed reading: the aggregate detection-AUC lift is *not* statistically robust ($+0.024 \pm 0.026$, reversing on one of five seeds, $p \approx 0.11$), whereas the per-row chain-recovery lift *is* ($+0.050 \pm 0.023$, positive on all five, $p \approx 0.008$). The multi-seed sweep also reproduced the iter-16 bimodal basin structure, confirming the cross-seed AUC variance is L-BFGS basin-selection rather than measurement noise. Tier 3 (Xuancheng), by contrast, remains a single day-pool (Apr 17–21): a multi-pool replication is the analogous outstanding confirmation there, a compute-cost trade-off rather than a methodological barrier.

L-BFGS basin selection on features=real. The Xuancheng pooled `features=real` rush-chain fit hit final loss 1477, against the `features=none` fit’s loss 4.156 on the same data. The same iter-15 multi-modality of L-BFGS that motivated the paper’s natural-gradient choice is the most likely explanation. We patched the multistart variant for Xuancheng (`score_seed_big_multistart.py`) but did not run it in the time available. This means the small features lift on the pooled Xuancheng pool (section 7.5) cannot be cleanly distinguished from a basin artefact. A multistart $K \geq 5$ run would resolve this in ~ 4 hours of wall-clock; flagged as immediate future work.

The hitting-rate ($f + g$) extension does not transfer to SUMO scale. The synthetic experiment of Tier 1 uses analytical (noise-free) g ; the SUMO and Xuancheng datasets produce empirical g with 84–89% of cells at the floor under any reasonable training-data size. The L-BFGS optimiser on the corresponding under-constrained landscape line-search-backtracks indefinitely. Under the simpler f -only loss the framework remains tractable. This is documented as a real — and surprising — limitation: the headline feature of the original paper transfers to synthetic but not to realistic-scale urban traffic data. The paper’s natural-gradient optimiser [1] §4.1 is the obvious algorithmic fix that we did not pursue here.

8.3 Social, environmental, technological implications

Transport-policy modelling. If externally-influenced routing constitutes even a few percent of urban traffic flow, demand models that assume revealed-preference inputs reflect intrinsic preferences are systematically misspecified. A method that quantifies and filters the externally-influenced fraction is a methodological prerequisite for any policy intervention (congestion pricing, infrastructure investment, route-recommendation regulation) whose evaluation rests on preference-revealed data.

Privacy and data access. The most-restrictive privacy control among the three real-world AVI datasets we surveyed is also the one that gives the strongest scientific guarantees: the Padua deposit [13] releases only aggregated flow statistics. The least-restrictive is the Ma *et al.* Xuancheng release [2] which publishes joinable per-trip records with anonymised IDs — enabling the per-trajectory detector this work relies on, but at the cost of stronger reidentification surface area. Methodologies that can validate per-trajectory detection from aggregate-only data would shift this trade-off favourably.

Sat-navigation regulation. The current sat-navigation recommendation pipeline is unregulated in most jurisdictions; the welfare effects of system-wide rerouting around congestion are contested. A practical detector that can statistically attribute trip fractions to sat-navigation guidance would underpin both academic welfare analyses and possible regulatory disclosure requirements.

8.4 Future work

1. **Multistart on features=real.** Highest expected information per hour of compute; resolves the open question of whether the small Xuancheng features lift is a basin artefact.
2. **Multi-seed Tier-2 confirmation.** Run the Tier-2 pipeline on the 5 pre-built SUMO seeds and report AUC_{fit} with confidence intervals. Required for Tier-1 reporting precision in the eventual paper writeup.
3. **Trajectory-level feature augmentation.** The strict 1-step Markov LR detector ignores trip-level features (length, mean speed, edge-class composition, dwell time at congested edges). A logistic-regression layer on top of $\Lambda(\tau)$ that adds those features would lift detection AUC further at the cost of a slightly weaker chain-recovery claim. Worth doing as a clearly-labelled “deployment-oriented” extension.
4. **Natural-gradient or LM optimisation.** Per section 6.6, the $f + g$ infeasibility is an optimiser-quality problem, not a method-class problem. The paper’s §4.1 natural-gradient descent or a Levenberg–Marquardt formulation against the residual Jacobian (iter-17 of the project explored this; abandoned at $d = 8,392$ for memory cost) would likely restore $f + g$ tractability and enable an $f + g$ -with-features re-validation.

5. **OD-conditional fitting on Xuancheng.** The OD-rebalancing extension matches the marginal ($\text{zone}_o, \text{zone}_d$) distribution. A stronger control fits per-cell chains and aggregates the per-cell contrasts; smaller sample sizes per cell but cleaner methodologically. The loader infrastructure required already exists.

9 Project Management and Reflection

This section addresses the *Project Management* and *Reflection* learning outcomes of the module assessment criteria [3].

9.1 Project structure and milestone management

The project was scoped at the outset as a three-tier validation pipeline (synthetic, simulated, real-world) extending a single source paper [1]. This top-down decomposition served well as a stable organising structure: each tier had clear inputs, outputs, and acceptance criteria, and the natural sequencing (upstream validation locks before downstream tiers depend on it) gave a clean de-risking order.

Internal milestone management was driven by a running `research_logbook.md` file that holds twenty “iter- N ” entries spanning the full project. Each iter documents one experimental cycle: hypothesis, setup, result, honest read, and next action. Two practices that worked particularly well:

- **The auto-memory file as an *epistemic* index.** A separate file (`MEMORY.md` with links to per-finding markdown records) was used for “things I now believe and the evidence for them” as opposed to “things I did and when”. Concretely, the file records statements such as “empirical g on SUMO is sparse to the point of being uninformative for the optimiser” with a pointer to the specific iter that established it. This separation made it possible to quickly distinguish “a working assumption that was right when I made it but might now be stale” from “a finding I have evidence for”.
- **Audit-before-extend.** A guiding rule, learned from iter-9’s retraction of an earlier headline number, was to audit the upstream implementation before extending it. Two specific audits saved time: (i) the gradient finite-difference check at $\sim 10^{-9}$ relative error confirmed the analytic gradient derivation before any sweep was run on features, avoiding the trap of attributing fit behaviour to a buggy derivative; (ii) the iter-18 contrast-correlation diagnostic established that AUC alone is an unreliable signal of chain quality, which materially altered how the Phase 4 result was framed.

Two practices that did not work as well are worth recording explicitly:

- **Underestimating the compute scaling of the $f + g$ loss.** The first $f + g$ sweep at SUMO scale was projected at ~ 1 hour and killed at 9.5 hours. The retry at smaller $|X_o|$ and `maxiter` was projected at ~ 30 minutes and killed at 3 hours. Two iterations were burned before the structural diagnosis (the hitting-rate gradient is $O(|X_o| \cdot n^3)$ and the 84%-floor loss landscape is degenerate at this scale) was reached. In future I would profile a single iteration before committing to a sweep at unfamiliar configuration.
- **Late identification of the OD-mix confound.** The Tier-3 rush-vs-off-peak proxy was scoped before the methodological cost of the OD-mix confound was clearly understood. The OD-rebalancing extension — the methodologically correct fix — was added in the final week, and as of writing the matched-pool experiment is still in flight. Earlier scoping would have caught this; the lesson is that “proxy weakness” should be a structured pre-experiment audit item, not an ad-hoc discussion item.

9.2 Benchmarking against the example dissertations

I read three example MEng Design Engineering dissertations from previous years to calibrate quality expectations. The benchmarks I took from each:

- **Aditya Munot, “Optimal Transport for Complex Sharing Economies with QoS Guarantees”**: the notation table at section 1 that gives readers a clean reference for the math. I adopted this convention in section 1. The clean separation of theoretical framework, scenario application, and benchmarking-against-state-of-the-art also informed how I structured the validation tiers.
- **Nora Luo, [thesis title placeholder]**: the discipline of foregrounding limitations in the discussion rather than burying them, and of writing the conclusion as a quantified claim ($X = Y$ % within tolerance Z) rather than a narrative. I have followed this style in section 8 where every claim is backed by a number.
- **Antanas Zilinskas, [thesis title placeholder]**: the use of figure-paired tables (e.g. a small numerical summary alongside a curve) rather than relying on either alone. I have not yet adopted this fully but the figure scaffolding is in place for the eventual revision.

By the assessment criteria of [3], the strengths I would foreground are: contextual investigation (a critical engagement with three real-world AVI datasets, only one of which is actually usable for per-trajectory work, and an explicit positioning against the source paper and rerouting-detection prior art); validation (three independent validation tiers, each with its own honest caveats and a quantified ceiling); and reflection (this section, and the honest framing of the $f + g$ infeasibility and the multistart question as future-work items rather than swept under the rug). The weaknesses are: single-seed Tier-2 and Tier-3 numbers, and the absence of multistart confirmation on `features=real`.

9.3 Personal skill development

The technical skills I leaned on heavily and improved in: deriving and finite-difference-verifying analytic gradients (Eqs. 12, 15, 17–18 of the source paper); writing software with a clean separation between the inverter’s mathematical core (in `morimura.py`, kept faithful to the paper) and the application-specific loaders that surround it (`sumo_to_phase3.py`, `real_data/load_xuancheng.py`); and writing a logbook that records both the successful and the unsuccessful experimental cycles in enough detail that I could critique my own past decisions a month later.

The non-technical skill I would name as the one that most shaped the project’s outcome is *honest framing of negative results*. The project did not produce the headline “real-world detector with a high AUC and clean chain recovery” that the SUMO Tier-2 result would have predicted. Instead it produced: a controlled-validation result (~ 0.71 AUC, 69% gap closure on labelled SUMO), an empirically-verified data ceiling (the 2-step refutation closes a tempting line of work), and an honest real-world result (~ 0.53 AUC on a behavioural proxy without ground-truth labels) that quantifies the gap between simulated and deployed detectability. Each of these is publishable, in part because they were framed honestly — but the temptation in week 30 of the project to switch the real-world contrast to a stronger but methodologically incorrect proxy (holiday-vs-normal) had to be resisted. Resisting that temptation in real time, against the prospect of a higher headline number, is the skill I most want to carry forward.

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A Repository structure and reproducibility

The codebase is organised as follows.

morimura.py The inverter core: `Inverter` class with softmax-parametric chain, analytic gradients (local + globals), λ -cross-validated fit, and the synthetic §6.1 reproduction driver.

congestion_filter.py Tier-2-precursor experiments (Phase 2 of the project’s internal phasing): snapshot-level two-regime synthesis and detection.

per_car_detector.py Tier-2 per-trajectory log-likelihood-ratio detector.

sumo_validation/ The SUMO-side machinery: `sumo_to_phase3.py` (network builder, trip reader, feature extractor), `score_seed_big.py` and `contrast_correlation.py` (scorers, with the `-dataset {sumo,xuancheng}` switch), `two_step_empirical.py` (the diagnostic of section 6.4), and `fd_check_features.py` (the FD gradient gate).

real_data/ The Xuancheng tier: the unmodified `xuancheng.net.xml` from [2], the daily JSON trip files, and `load_xuancheng.py` which provides the loader, the three regime-split helpers, and the OD-rebalancing extension.

papers/ Source PDFs of all cited works.

research_logbook.md The day-by-day project log, twenty iter- N entries.

B Use of generative AI

In accordance with the module’s declaration requirement, generative AI was used in this project as follows. An LLM-based code assistant (Claude) was used as a pair-programming aid throughout the implementation phase: drafting code (with the author’s careful review of every line before commit), writing unit-test scaffolding, deriving analytic gradient formulas alongside the author, and stress-testing arguments before they made it into the writeup. The same assistant contributed to drafts of this report, working from the author’s running `research_logbook.md` and the experimental result logs; the author edited every section for accuracy and final framing. All experimental decisions, all interpretations of results, and the choice of what counts as an honest finding remained with the author. Specifically the decision to refuse the holiday-vs-normal regime proxy in favour of the OD-matched rush-vs-off-peak contrast (sections 7.3 and 9.3) was made against the assistant’s initial suggestion, on methodological grounds.

C Build and reproducibility

The project is reproducible from the repository via the README, with the following minimum environment:

```
# Python 3.14, see project README
python3 -m venv .venv
.venv/bin/python -m pip install numpy scipy matplotlib

# Tier 1: synthetic Morimura reproduction (~20 s)
.venv/bin/python morimura.py

# Tier 2: SUMO experiments (requires SUMO_HOME set)
.venv/bin/python sumo_validation/contrast_correlation.py \
    --seed 42 --features real --T 40 --maxiter 1500

# Tier 3: Xuancheng experiments (requires real_data/ populated)
.venv/bin/python sumo_validation/contrast_correlation.py \
    --dataset xuancheng \
    --day 2023-04-17,2023-04-18,2023-04-19,2023-04-20,2023-04-21 \
```

```
--regime_split rush_offpeak --od_match 8 \  
--features real --T 10 --maxiter 1500
```